Modeling Pit Stop Strategy in Formula 1: A Machine Learning Approach

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***Abstract*—Pit stop decisions are a central part of Formula 1 race strategy and are influenced by tire condition, race progress, and driver performance. The goal of this project is to predict whether a driver will pit on the next lap using structured lap-level data. We compare logistic regression and a simple neural network, and evaluate them using precision, recall, F1-score, and ROC-AUC. We expect features related to tire life, lap-time changes, and race progression to improve prediction performance. The proposed Streamlit application will provide an interactive tool for students, Formula 1 fans, and analysts to explore how pit stop decisions change under different race conditions.**

I. Introduction

A. Problem and Importance

In Formula 1, race outcomes are strongly influenced by in-race strategy, and pit stop timing is one of the most important decisions. Teams must balance undercut and overcut opportunities with tire degradation and track position. A poorly timed pit stop can lead to position losses that are difficult to recover, especially on tracks where overtaking is limited. For this reason, understanding pit stop decisions is an important race-strategy problem.

B. Overview of Existing Work

Traditional approaches to race strategy rely on expert knowledge, simulation, and predefined rules [1]. Although these methods can capture general patterns, they often struggle to adapt to changing race conditions. With the availability of lap-level data, pit stop decisions can now be studied more systematically. Recent work has explored data-driven methods for performance analysis and race strategy modeling [2], but many approaches still focus on overall race outcomes rather than in-race decisions.

C. Gap in Existing Work

Despite the availability of detailed data, there is limited work that models pit stop decisions at the lap level. Existing approaches do not fully capture how multiple factors, such as tire life, lap time changes, and race progression, jointly influence decision making. This creates a gap between the available data and its use in understanding race strategy.

D. Proposed Approach

In this project, we predict whether a driver will pit on the next lap. We use logistic regression as a baseline and a simple neural network to capture nonlinear relationships. We evaluate performance using precision, recall, F1-score, and ROC-AUC to account for class imbalance. We also build a Streamlit application that allows users, including students, fans, and analysts, to input race conditions and view predicted pit decisions.

E. Broader Impact

This project provides a data-driven perspective on race strategy grounded in real race dynamics. It supports more systematic analysis of pit decisions and illustrates how machine learning can be used to study sequential decision making in real-world settings.

II. Background

A. Comparisons to Prior Work

Formula 1 race strategy has traditionally been studied using analytical models and simulation-based methods [1]. These approaches focused on tire degradation and pit stop timing under simplified assumptions, and later simulations incorporated factors such as track position and traffic. Although

they improved realism, they still relied heavily on predefined rules. More recent studies have explored learning-based methods for race strategy decisions [2]. However, most prior work still emphasizes overall race outcomes or strategy optimization rather than directly modeling decisions at each lap.

B. Common Knowledge and Discussion

There is ongoing discussion about how pit stop decisions are made during a race. Some approaches emphasize tire degradation, while others place greater importance on track position, traffic, and overall race context [1], [2]. In practice, these factors interact and can change quickly as the race evolves. This makes pit stop timing difficult to model using a single rule or assumption, and there is no clear agreement on which variables should be prioritized.

C. Knowledge Gaps and Proposed Contribution

Based on these limitations, there is a gap in modeling pit stop decisions at the lap level using real race data. Existing approaches either rely on simulation or focus on aggregate outcomes rather than decisions at each time step. Our approach addresses this gap by using lap-level data to predict pit decisions. We use logistic regression as a baseline and a simple neural network to capture both linear and nonlinear relationships. We expect this approach to improve understanding of how multiple race factors interact and to support more systematic analysis of pit stop decisions.

III. End-to-End Machine Learning Pipeline

A. Offline Model Training and Evaluation

1) Data Collection, Exploration & Processing: We use a lap-level Formula 1 race dataset where each row represents a single lap completed by a driver. The dataset contains numerical, categorical, and time-related features, including lap time, tire life, degradation measures, tire compound, and driver identity. Although the dataset has sequential structure, we formulate the task as a lap-level binary classification problem and model each lap independently.

The dataset is publicly available on Kaggle and is a processed, feature-engineered version of publicly accessible Formula 1 telemetry and timing data collected using the FastF1 Python library. It contains 101,371 lap-level observations across 28 races, 31 drivers, and the 2022–2025 seasons. We use this dataset to predict `PitNextLap`, which indicates whether a driver will pit on the next lap. This target variable is directly provided in the dataset.

The dataset was constructed by loading race sessions through FastF1, extracting lap-level variables, converting time-based values into numerical form, and engineering additional features such as lap time delta, cumulative tire degradation, normalized tire life, race progress, and position change. Pit-stop events were identified from tire-life resets, and incomplete or inconsistent entries were removed during data cleaning.

For data exploration, we focus on two visualizations that are directly related to the prediction task. First, we use a bar chart of `PitNextLap` to examine the class distribution and identify class imbalance.

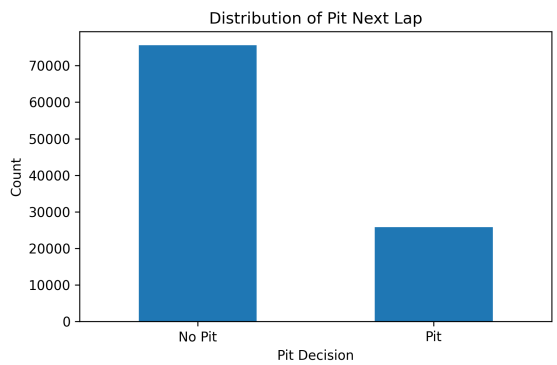

Fig. 1. Class distribution of `PitNextLap`. The figure shows the number of lap-level observations in the non-pit and pit-next-lap classes, highlighting class imbalance in the dataset.

As shown in Figure 1, the dataset is imbalanced across the two classes. Second, we use a boxplot of `TyreLife` grouped by `PitNextLap` to compare tire usage patterns between pit and non-pit laps.

Figure 2 shows that `TyreLife` is generally higher for laps that are followed by a pit stop. Together, these plots help assess whether tire life is a useful predictor of pit-stop behavior.

For preprocessing, we remove missing or invalid entries, normalize numerical variables, encode categorical variables into numerical form, and create additional features such as lagged lap time and

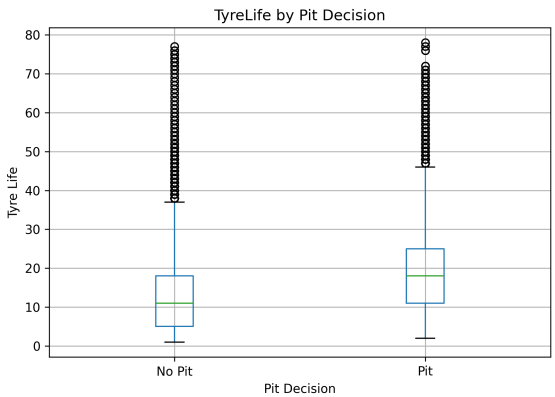

Fig. 2. Distribution of `TyreLife` grouped by `PitNextLap`. The boxplot compares tire usage patterns between pit and non-pit laps and shows that laps followed by a pit stop tend to have higher tire life values.

recent performance trends. These steps make the dataset suitable for model training while preserving the race context needed for prediction.

2) Methods and Model Training: We formulate this problem as a binary classification task. Let $x \in R^d$ denote the lap-level feature vector and let $y \in \{0, 1\}$ denote the target label, where $y = 1$ indicates a pit stop on the next lap.

We use two models. The first is logistic regression, which serves as an interpretable baseline. Its predicted probability is

$$P(y = 1 \mid x) = \sigma(w^\top x + b),$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$. The model is trained using binary cross-entropy loss:

$$\mathcal{L} = -\frac{1}{n} \sum_{i=1}^{n} \left[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \right].$$

The second model is a simple neural network implemented in NumPy with one hidden layer:

$$h = \phi(W_1 x + b_1), \qquad \hat{y} = \sigma(W_2 h + b_2),$$

where $\phi(\cdot)$ is a nonlinear activation function. This model captures nonlinear relationships among race variables.

The inputs to both models include lap-level features such as `TyreLife`, `LapTime_Delta`, `RaceProgress`, degradation-related variables, and encoded categorical variables such as `Compound`, `Driver`, and `Race`. The output is a binary prediction for `PitNextLap`. Both models are implemented without high-level machine learning libraries, using only NumPy and Pandas.

These models are well-suited for this problem because they balance interpretability and flexibility. Logistic regression provides a clear baseline, while the neural network can capture more complex feature interactions. Comparing them allows us to test whether additional model complexity improves pit-stop prediction at the lap level.

3) Model Evaluation: We evaluate model performance using precision, recall, F1-score, and ROC-AUC. These metrics are appropriate because pit stop events are relatively rare, so accuracy alone would not provide meaningful evaluation. Precision, recall, and F1-score are defined as

$$\text{Precision} = \frac{TP}{TP + FP},$$

$$\text{Recall} = \frac{TP}{TP + FN},$$

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

Precision measures how many predicted pit-stop events are correct, recall measures how many true pit-stop events are identified, and F1-score balances the two. ROC-AUC evaluates classifier performance across decision thresholds.

We split the dataset into training and testing sets using a 70/30 split. A validation split is created from the training data for model selection and hyperparameter tuning. We compare logistic regression and the neural network under the same data split and tune hyperparameters such as learning rate, regularization strength, and hidden-layer size.

To reduce overfitting, we use regularization in logistic regression, limit the complexity of the neural network, and monitor performance on the training and validation sets. Final results are reported on the held-out test set using the same evaluation procedure for both models.

4) Model Deployment: The final model will be selected based on its performance on the held-out test set, with particular attention to precision, recall, F1-score, and ROC-AUC. We will compare the trade-offs between interpretability and predictive accuracy when choosing the deployed model.

If logistic regression achieves competitive performance, it may be preferred because it provides a simpler and more interpretable decision rule. If the neural network produces substantially better predictive performance, it will be selected despite its higher complexity. The selected model will be saved and integrated into the Streamlit application, where users can enter race conditions and receive real-time pit-stop predictions.

B. Front-End Application Using Streamlit

1) Target Users: The application is designed for users interested in race strategy analysis, including students, Formula 1 fans, and other non-technical users who want to explore how pit stop decisions are made.

2) User Interface Design: The application is designed as a simple and interactive dashboard. The layout includes an input panel on the left, a prediction output area on the upper right, and two analysis panels below for sensitivity analysis and feature-level insights. As shown in Figure 3, the interface combines user inputs, model predictions, and interactive analysis in a single dashboard.

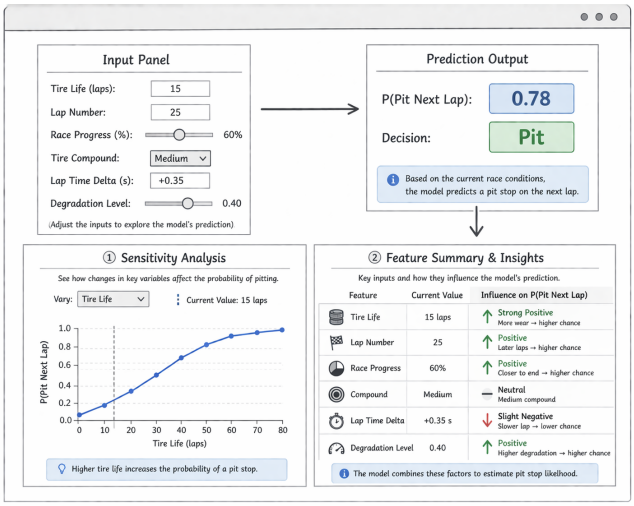

Fig. 3. Proposed Streamlit interface for interactive pit-stop prediction. The layout includes an input panel, a prediction output area, a sensitivity analysis panel, and a feature summary panel.

3) User Inputs: Users enter race conditions through sliders and dropdown menus, including tire life, lap number, race progress, tire compound, lap time change, and degradation-related variables. Each input corresponds to a model feature and is restricted to realistic ranges.

4) Outputs and Visualization: The application outputs the predicted probability of a pit stop on the next lap and a binary prediction based on a decision threshold. It also provides a sensitivity analysis panel and a feature summary panel to help users interpret how key inputs influence the prediction.

5) System Integration and Use Case: The trained model is saved and loaded into the Streamlit application for real-time prediction. User inputs are converted into the same feature format used during training and passed to the model to generate a prediction. A typical use case is that a user adjusts race conditions, such as tire life or race progress, and observes how the predicted pit-stop probability changes.

IV. Risk and Mitigation

One anticipated challenge is class imbalance, since pit stop events are less frequent than non-pit laps. To mitigate this issue, we will monitor model performance on both classes and use precision, recall, and F1-score instead of relying only on accuracy. We will also consider threshold adjustment if needed.

Another challenge is potential overfitting, especially for the neural network model. To reduce this risk, we will limit model complexity, apply regularization, and monitor performance on the validation set during development.

A third challenge is implementing the models without high-level machine learning libraries. To address this, we will use NumPy, test each component step by step, and validate intermediate outputs to ensure correctness.

V. Expected Outcomes

We expect that features related to tire life, lap time changes, and race progress will have meaningful predictive power for pit stop decisions. We also expect the neural network to perform competitively with logistic regression and potentially achieve stronger predictive performance if nonlinear interactions among race variables are important. Finally, we expect that the Streamlit application will provide a clear and interactive way to explore pit stop predictions.

VI. Team Member Contribution

Technical Contributions: He Sun is responsible for data preprocessing and feature engineering. Yilin Wang focuses on implementing the logistic regression model and evaluating its performance. Weifan Wu is responsible for building the neural network in NumPy and tuning model parameters. Peter Ye is responsible for developing the Streamlit application and integrating the trained model into the user interface.

Writing Contributions: He Sun writes the Introduction and Background sections. Yilin Wang writes the Machine Learning Pipeline section. Weifan Wu writes the Model Evaluation and Streamlit sections. Peter Ye writes the Risk and Expected Outcomes sections and reviews the full report for consistency.

References

[1] A. Heilmeier, M. Graf, and M. Lienkamp, "A race simulation for strategy decisions in circuit motorsports," in *2018 IEEE Intelligent Transportation Systems Conference (ITSC)*, 2018, pp. 2986–2993.

[2] A. Heilmeier, A. Thomaser, M. Graf, and J. Betz, "Virtual strategy engineer: Using artificial neural networks for making race strategy decisions in circuit motorsport," *Applied Sciences*, vol. 10, no. 21, p. 7805, 2020.